Math 55 Second Midterm Exam, Prof. Srivastava
November 12, 2024, 5:10pm–6:30pm, 150 Wheeler.

Name: ______________________________

SID: ______________________________

GSI: ______________________________

Name of the student to your left: ______________________________
Name of the student to your right: ______________________________

Instructions: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. **Do not unstaple the exam.** Write your name and SID on every page. Show your work — numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.
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Sign here: ______________________________

Question	Points
1	8
2	6
3	6
4	8
5	9
6	6
7	7
Total:	50

Do not turn over this page until your instructor tells you to do so.

Name and SID: ______________________________

1. Select true or false (by filling in the circle) for each of the following. There is no need to provide an explanation.

   (a) (2 points) If $n \geq 2$ and $1 \leq r \leq n-1$, then $\binom{n}{r}$ is divisible by n.
   ◯ True ◯ False

   > **Solution:** False. The easiest counterexample is to consider $r = 2$, in which case $\binom{n}{2} = n(n-1)/2$. Taking $n = 4$ shows that $\binom{n}{2} = 6$, which is not divisible by 4.

   (b) (2 points) If $n \geq 1$ and $0 \leq a \leq b \leq n$ then $\binom{n}{a} \leq \binom{n}{b}$.
   ◯ True ◯ False

   > **Solution:** False. If $a = n-1$ and $b = n$ then $\binom{n}{a} = n$ and $\binom{n}{b} = 1$, which is less than n for $n \geq 2$.

   (c) (2 points) If T is a tree then T has exactly two $2-$colorings.
   ◯ True ◯ False

   > **Solution:** True. The proof is by induction on the number of vertices. The base case holds since a single vertex has two $2-$colorings. For the inductive step, assume the claim is true for trees with at most k vertices. Let T be a tree with $k+1$ vertices. Choose a leaf x of T and consider the subgraph $T' - x$, which is also a tree as shown in class. Let y be the vertex of T adjacent to x. Notice that every $2-$coloring f of T is uniquely specified by its restriction f' to T': the color $f(x)$ must be the opposite of $f(y) = f'(y)$. Thus, the number of $2-$colorings of T is equal to the number of $2-$colorings of T', which by the inductive hypothesis is equal to 2.

   (d) (2 points) If $G = (V, E)$ is a simple graph then the number of distinct subgraphs of G is at least $2^{|E|}$.
   ◯ True ◯ False

   > **Solution:** True. We can define a subgraph $H = (V, F)$ of G by deciding for each edge $e \in E$ whether or not $e \in F$. Moreover, all of these subgraphs are distinct since they have distinct sets of edges. By the product rule, the number of ways to choose such a subgraph is $2^{|E|}$, so the total number of distinct subgraphs is at least $2^{|E|}$.

   [Scratch Space Below]

2. (6 points) Prove by induction that for every integer $n \geq 4$,

$$2^n \leq n!.$$

Solution: Let $P(n)$ be the statement that $2^n \leq n!$.

Base Case: $P(4)$ is true because $2^4 = 16$ and $4! = 24$.

Inductive Step: Assume $k \geq 4$ and assume $P(k)$. We will show $P(k+1)$. Observe that

$$2^{k+1} = 2 \cdot 2^k \leq 2 \cdot k!$$

by $P(k)$. Moreover, as $2 \leq k+1$ we have $2 \cdot k! \leq (k+1) \cdot k!$. Combining this with the previous inequality yields

$$2^{k+1} \leq 2 \cdot k! \leq (k+1)!,$$

as desired.

[Scratch Space Below]

3. (6 points) Consider the function $f : \mathbb{Z}_+ \to \mathbb{Z}_+$ recursively defined by $f(1) = 1$

$$f(n) = f(n-1) + 2n - 1, \qquad n \geq 2.$$

Show that for all $n \in \mathbb{Z}_+$, $f(n)$ is not prime.

Solution: We will show a stronger statement. We proceed by induction. Let $P(n)$ be the proposition "$f(n) = n^2$". We will show that $P(n)$ is true for all $n \geq 1$, from which we can conclude that $f(n)$ is not prime since n^2 is not prime for any $n \geq 1$.

Base case: $f(1) = 1^2$.

Inductive Step: Let $k \geq 1$ and assume $P(k)$. Observe that

$$f(k+1) = f(k) + 2(k+1) - 1 = k^2 + 2k + 1 = (k+1)^2,$$

so $P(k+1)$ is established, as desired.

This is an example of inductive loading. It was necessary to strengthen the inductive hypothesis from "$f(n)$ is not prime" to $P(n)$ as above, since there is no clear way to use the fact that $f(k)$ is not prime in the inductive step. The most natural way to come up with this hypothesis would be to compute $f(1), f(2), f(3)$ and notice the pattern that they are squares.

4. (a) (4 points) Prove that if T is a tree with n vertices, then T must have exactly $n-1$ edges. (You may not just cite that this was proven in class, but you may use other properties of trees which we proved).

Solution: We proceed by induction on the number of vertices of the tree. Let $P(n)$ be the proposition "If T is a tree with n vertices, then T must have exactly $n-1$ edges."

Base Case. When $n = 1$ the claim is true.

Inductive Step. Let T be a tree on $k+1$ vertices. By a theorem in class, T has at least one leaf vertex x. Let y be the unique vertex adjacent to x in T and let $T' = T - x$. Notice that T' is still connected since every pair of vertices in T' is connected by a path which does not visit x; to see that, observe that every pair a, b of distinct vertices in T' is connected by a simple path, and no simple path in T can visit x as it would traverse y twice. Thus, by the inductive hypothesis T' has $k-1$ edges. T has exactly one more edge than T', namely xy, so T has k edges, as desired.

This is Theorem 16.3 from the graph theory notes.

(b) (4 points) Prove that if $n \geq 1$ and $1 \leq k \leq n$, and G is an acyclic graph with n vertices and exactly k connected components, then G must have exactly $n-k$ edges. You may use the statement of part (a), even if you did not prove it.

Solution: Assume $n \geq 1$ and $1 \leq k \leq n$. Assume G is acyclic with n vertices and k connected components. Let $G_1, \ldots G_k$ be the connected components of G. Suppose G_i has n_i vertices for $i = 1, \ldots k$. Observe that each G_i is acyclic, since if G_i contained a cycle as a subgraph then G would contain a cycle as a subgraph. Since each G_i is also connected, each G_i is a tree. By part (a), the number of edges of G_i is exactly $n_i - 1$. Since G is a disjoint union of $G_1, \ldots, G_k$, the total number of edges of G is

$$\sum_{i=1}^{k}(n_i - 1) = (\sum_{i=1}^{k} n_i) - k = n - k,$$

as desired.

5. Recall that the degree sequence of a graph is a list of the degrees of its vertices in nondecreasing order. Give examples of graphs with the following properties, or prove that no such graph can exist.

   (a) (3 points) A simple graph with degree sequence $2, 2, 1, 1$.

   Solution: Let $V = \{1, 2, 3, 4\}$ and $E = \{12, 23, 34\}$. Then vertices 2 and 3 have degree 1 and the remaaining vertices have degree 1.

   (b) (3 points) A tree with 5 vertices which has an Eulerian circuit.

   Solution: No such tree exists. This is because every tree is connected and has leaf, i.e., a degree 1 vertex. But a graph has an Eulerian circuit iff it is connected and all of its vertices have even degree.

(c) (3 points) A simple graph with 4 vertices and 5 edges which is $2-$colorable.

Solution: No such graph exists. One way to see this is: a simple graph G on 4 vertices with 5 edges is equal to the complete graph on 4 vertices $\{a, b, c, d\}$ (which has six edges) minus an edge. Without loss of generality you can assume that the edge is ab (by possibly relabeling the vertices), so $G = (\{a, b, c, d\}, \{bc, cd, bd, ad, ac\}$. Notice that G has a cycle of length 3, namely bc, cd, bd. Since a graph is $2-$colorable iff it has no odd circuit, G cannot be $2-$colorable.

6. (6 points) Give a combinatorial proof that $\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}$.

Solution: Let us count the number of nomempty subsets of $[n]$ with one marked element, i.e.,

$$A = \{(S, x) : S \subseteq [n], x \in S\}.$$

First, an element of A may be uniquely specified by the following process: (1) Choose an integer $1 \le k \le n$. (2) Choose a $k-$subset S of $[n]$ ($\binom{n}{k}$ ways). (3) Choose an element x of S (k ways). Given a choice of k in step (1), the number of ways of executing steps (2,3) is $k\binom{n}{k}$, by the product rule. By the sum rule, the total number of ways of choosing elements of A is therefore

$$\sum_{k=1}^{n} k\binom{n}{k}.$$

On the other hand, we may also uniquely specify an element of A as follows: (i) Choose the marked element $x \in [n]$ (n ways). (ii) Choose the remaining elements of S, i.e., a subset $T \subseteq [n] - \{x\}$ (2^{n-1} ways). (iii) Let $S = T \cup \{x\}$. By the product rule, the number of ways of choosing an element of A is $n2^{n-1}$.

Thus, we have

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1},$$

as desired.

This was by far the hardest question on the exam. It was difficult because it is not obvious what either side is counting. The biggest clue is the left hand side: $\binom{n}{k}$ is counting $k-$subsets of $[n]$ by definition, and each such subset has k elements so it is natural to consider choosing one of them.

7. (a) (3 points) What is the cardinality of the set:

$$S = \{(x_1, x_2, x_3) : x_1, x_2, x_3 \text{ are positive integers and } x_1 + x_2 + x_3 = 5\}?$$

Justify your answer.

Solution: Each element of S corresponds to a placement of 5 unlabeled tokens into 3 labeled boxes such that each box has at least one token. Such a placement is specified by the following process (i) Place a token in each box (1 way) (ii) Place the remaining 2 tokens in the 3 boxes ($\binom{3+2-1}{3-1} = \binom{4}{2} = 6$ ways). Thus, the total number of such placements is 6, and $|S| = 6$.

(b) (4 points) What is the cardinality of the set:

$$T = \{(x_1, x_2, x_3) : x_1, x_2, x_3 \text{ are even positive integers and } x_1 + x_2 + x_3 \leq 11\}?$$

Justify your answer.

Solution: Since x_1, x_2, x_3 are even, their sum cannot equal 11, and we have

$$T = \{(x_1, x_2, x_3) : x_1, x_2, x_3 \text{ are even positive integers and } x_1 + x_2 + x_3 \leq 10\}.$$

Moreover, writing $x_i = 2y_i$ for each even positive even integer x_i, the set T is in bijection with

$$T' = \{(y_1, y_2, y_3) : y_1, y_2, y_3 \text{ are positive integers and } y_1 + y_2 + y_3 \leq 5\}.$$

Since $y_1, y_2, y_3 > 0$, the possible values of $y_1 + y_2 + y_3$ for $(y_1, y_2, y_3) \in T$ are $3, 4, 5$.

In the first case, there is only one way to choose $(y_1, y_2, y_3) = (1, 1, 1)$.

The second case corresponds to placing one unlabeled token into one of three labeled boxes, which can be done in 3 ways.

The final case was solved in part (a).

By the sum rule, the total number of ways is $1 + 3 + 6 = 10$.

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[Scratch Paper]

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Graph Theory Definitions

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a finite multiset of 2−element subsets of V, called **edges**. If E has repeated elements G is called a **multigraph**, otherwise it is called a **simple graph**. Two vertices $x, y \in V$ are **adjacent** if $\{x, y\} \in E$. An edge $e = \{x, y\}$ is said to be **incident with** x and y. The **degree** of a vertex x is the number of edges incident with it.

A **path** of length k in G is an alternating sequence of vertices and edges

$$x_0, e_1, x_1, e_2, \ldots, x_{k-1}, e_k, x_k$$

where $e_i = \{x_{i-1}, x_i\} \in E$ for all $i = 1 \ldots k$. The path is said to **connect** x_0, x_k, **visit** the vertices $x_0, \ldots, x_k$, and **traverse** the edges $e_1, \ldots, e_k$. A path is called **simple** if it contains no repeated vertices. If $x_0 = x_k$ then the path is called a **circuit**. A graph is **connected** if it contains a path between every pair of distinct vertices.

A **k-coloring** of G is a function $f : V \to \{1, \ldots, k\}$ such that $f(x) \neq f(y)$ whenever x is adjacent to y. A graph is $k-$**colorable** if it has a k-coloring.

A graph $H = (W, F)$ is a **subgraph** of G if $W \subseteq V$ and $F \subseteq E$. If $W = V$ then H is called a **spanning** subgraph of G. A **connected component** of G is a maximal connected subgraph of G, i.e. a subgraph $H \subseteq G$ such that H is connected and adding any vertices or edges to H would make it disconnected.

A **cycle** is a circuit with no repeated edges and no repeated vertices, except the endpoints. A graph is called **acyclic** if it does not contain a cycle as a subgraph. A connected acyclic graph is called a **tree**.

An **Eulerian circuit** of a connected multigraph G is a circuit which traverses every edge of G exactly once.